

TALRAG: A Self-Learning Temporal-Adaptive RAG Framework

NGUYEN Quoc-Dat¹, Ngo-Ho Anh-Khoa¹, and Ngo-Ho Anh-Khoi^{1*}

¹ Faculty of Information Technology, Nam Can Tho University, 168 Nguyen Van Cu Street, An Binh Ward, Can Tho City, Vietnam
{dat223955@student.nctu.edu.vn, nhakhoa@nctu.edu.vn, nhakhoi@nctu.edu.vn}

Abstract. Retrieval-Augmented Generation (RAG) has become a popular method for reducing hallucinations in Large Language Models (LLMs). However, current RAG frameworks usually rely on static knowledge bases and basic semantic retrieval, which limits their ability to handle domain-specific reasoning and update with new information. This limitation is a major issue in historical question answering systems, where temporal context, causal links, and factual reliability are highly important. To address this, this paper proposes TALRAG, a self learning and temporal adaptive RAG framework built specifically for Vietnamese Historical Question Answering. The system combines adaptive retrieval using multiple scores, self learning guided by trust assessment, and a human-in-the-loop mechanism into a single architecture to keep the knowledge evolving. Unlike standard RAG models, TALRAG continuously updates its knowledge base through a learning pipeline guided by feedback that involves web information gathering, reliability checking, user validation, and automatic vector ingestion. To improve how the model reasons about history, we introduce an retrieval mechanism based on query intent that dynamically scores and combines semantic relevance, time proximity, and causal relationships based on what the user is asking. The whole process runs through a pipeline with multiple stages, including FAISS-based retrieval from multiple vector stores, LLM-based document grading, a web fallback for missing data, and automatic knowledge refinement. Additionally, our learning approach based on user feedback allows new external historical data to become trusted vector memory once validated by users, which improves future search quality and cuts down on hallucination risks. Our experimental results show that TALRAG clearly outperforms traditional static RAG architectures in retrieval robustness, answer relevance, and continuous knowledge adaptation. Overall, this proposed framework offers a scalable and reliable solution for domain-specific AI assistants that require trustworthy reasoning and the ability to learn over time.

Keywords: Retrieval-Augmented Generation (RAG); Self-Learning RAG; Human-in-the-Loop Learning; Temporal Adaptive Retrieval; Historical Question Answering; Knowledge Evolution; Large Language Models.

1 INTRODUCTION

In recent years, the rapid growth of Artificial Intelligence (AI) and digital information has increased the demand for question answering (QA) systems across many domains. Compared with traditional keyword-based search, modern QA systems provide more natural and context-aware access to information (Riloff & Thelen, 2000). The development of Transformer-based models (Vaswani et al., 2017), including BERT (Devlin et al., 2019) and GPT-4 (OpenAI, 2023), has further improved natural language understanding and generation.

Despite these advances, Large Language Models (LLMs) still suffer from hallucination, where generated outputs may contain unsupported or incorrect information (Ji et al., 2023). This issue is especially critical in historical question answering because historical answers require factual accuracy, temporal consistency, and reliable causal interpretation. Therefore, a historical QA system must not only retrieve relevant facts but also reason correctly about dates, dynasties, events, and cause and effect relationships. Retrieval-Augmented Generation (RAG) reduces hallucination by grounding LLM responses in external knowledge sources rather than relying only on model parameters (Lewis et al., 2020; Gao et al., 2024). However, conventional RAG still has limitations in historical QA because it often relies mainly on semantic similarity, lacks explicit temporal and causal reasoning, and usually operates on static knowledge bases that cannot learn from user feedback (Dhingra et al., 2022; Asai et al., 2023; Yan et al., 2024). To address these limitations, this paper introduces TALRAG, a RAG framework with self learning and temporal adaptation for Vietnamese historical question answering. Unlike static RAG models, TALRAG learns from validated user feedback and updates its vector memory through a self learning pipeline guided by trust assessment.

The rest of this paper is organized as follows. Section 2 reviews related work on RAG, adaptive retrieval, and self learning QA systems. Section 3 presents the proposed TALRAG framework. Section 4 describes the experimental setup and results. Section 5 concludes the paper and discusses future work.

2 RELATED WORK

The development of question answering (QA) systems has evolved from rule-based methods to neural and retrieval-augmented approaches. Early QA systems relied mainly on handcrafted rules and keyword matching, which provided controllability but limited contextual reasoning capability (Riloff & Thelen, 2000; Lehnert, 1977). The emergence of Transformer-based models significantly improved natural language understanding and generation (Vaswani et al., 2017). Pre-trained models such as BERT and GPT-4 further advanced contextual representation, reasoning, and conversational interaction (Devlin et al., 2019; OpenAI, 2023). However, LLMs still suffer from hallucination, where generated answers may contain unsupported or factually incorrect information (Ji et al., 2023). This limitation is especially critical in

historical QA, where factual accuracy, temporal consistency, and causal interpretation are essential.

Retrieval-Augmented Generation (RAG) addresses hallucination by grounding LLM outputs in external documents (Lewis et al., 2020). Dense retrieval methods such as DPR and Sentence-BERT improve semantic search, while FAISS enables efficient vector retrieval over large-scale document collections (Karpukhin et al., 2020; Reimers & Gurevych, 2019; Johnson et al., 2019). Recent surveys show that RAG has become an important approach for knowledge-intensive NLP tasks (Gao et al., 2024). Nevertheless, conventional RAG systems still face limitations in historical question answering. They often rely mainly on semantic similarity, which may retrieve documents that are linguistically related but temporally or causally inconsistent. Prior work on time-aware language models highlights the importance of temporal reasoning for knowledge-intensive tasks involving timelines and evolving facts (Dhingra et al., 2022).

Several studies have attempted to improve retrieval quality and answer reliability. BERT-based re-ranking improves passage ranking by refining the order of retrieved documents (Nogueira & Cho, 2019), while BEIR provides a heterogeneous benchmark for evaluating retrieval models across domains (Thakur et al., 2021). More recent frameworks such as Self-RAG and Corrective RAG introduce self-reflection or correction mechanisms to improve generation quality (Asai et al., 2023; Yan et al., 2024). GraphRAG further explores graph-based retrieval to capture global relationships among entities (Edge et al., 2024). However, these approaches still do not fully address the need for a transparent mechanism that continuously updates domain knowledge from validated user interaction history.

In the Vietnamese NLP domain, Nguyen and Nguyen (2020) introduced PhoNLP, which supports Vietnamese language processing and is relevant to historical QA because Vietnamese historical questions often contain named entities, dynastic names, temporal expressions, and culturally specific terminology. For the historical QA baseline, this study uses ItihashQA. ItihashQA is a retrieval-augmented conversational question answering system developed for the Bangladeshi historical context. Its released implementation follows a static RAG architecture, where retrieved historical contexts are passed to a language model for response generation. Since ItihashQA is specifically designed for historical question answering, this study adopts it as the baseline system and adapts its static RAG pipeline to the Vietnamese historical corpus for controlled comparison. Although the repository description also refers to the system as EkattorQA, this paper consistently uses the name ItihashQA according to the repository and work title.

Commercial document-grounded QA tools such as Google NotebookLM and Gemini Gems also demonstrate the practical value of retrieval-based assistants for document understanding and personalized knowledge access. These systems allow users to organize sources or create customized assistants for specific tasks. However, their retrieval mechanisms, evaluation procedures, and memory update strategies are not fully transparent for research reproduction. More importantly, they do not provide an explicit self learning pipeline guided by trust assessment, in which chat history

validated by users is converted into new vector memory. Therefore, although these tools are useful for document synthesis and personalized assistance, they are not directly equivalent to TALRAG’s knowledge evolution mechanism guided by feedback.

From the reviewed literature, three limitations remain clear. First, most RAG systems rely heavily on semantic similarity and do not explicitly optimize retrieval for temporal and causal reasoning, which are essential in historical QA. Second, many adaptive RAG frameworks improve retrieval or generation quality but still operate on static knowledge bases after deployment. Third, existing document-grounded chatbot systems do not provide a transparent research-oriented mechanism for converting user feedback and chat history into validated vector memory.

TALRAG addresses these gaps by combining retrieval based on query intent, temporal and causal scoring, document grading using an LLM, web fallback for missing information, and self learning guided by feedback. The key novelty of TALRAG is its ability to learn from user chat history. When the system encounters a low-confidence or unanswered query, the interaction becomes a learning candidate. After external knowledge acquisition, reliability verification, and positive user feedback, the validated knowledge is refined and ingested into the dynamic FAISS vector store. In this way, TALRAG does not simply answer questions from a fixed corpus; it continuously evolves its historical knowledge base through knowledge validation guided by feedback.

3 PROPOSED TALRAG FRAMEWORK

The main contribution of TALRAG is a self learning pipeline guided by trust assessment that converts low-confidence or unanswered queries into opportunities for knowledge expansion. Unlike standard RAG systems that rely on fixed vector databases, TALRAG uses validated user feedback and chat history to update its dynamic vector memory over time.

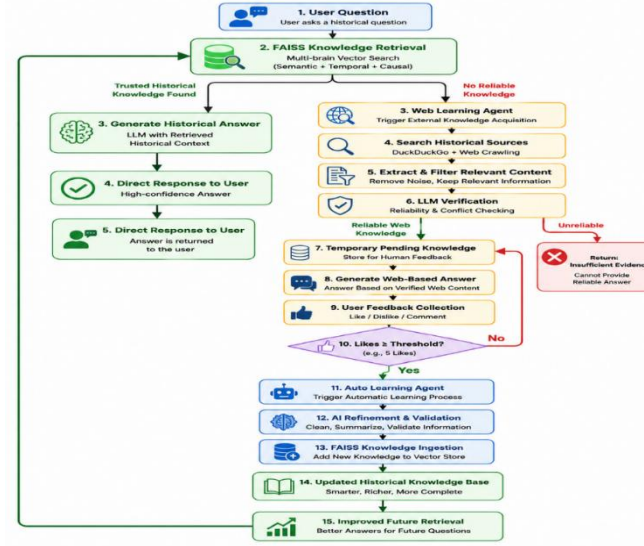


Fig. 1. self learning pipeline guided by trust assessment of TALRAG.

To formally describe the proposed workflow, Algorithm 1 presents the main inference pipeline, including intent classification, temporal and causal retrieval, document grading, and answer generation. Algorithm 2 describes the learning pipeline based on user feedback hoặc self learning pipeline, where retrieval failures are converted into learning candidates through web acquisition, reliability verification, user feedback, and FAISS ingestion.

Algorithm 1. TALRAG inference pipeline

Input: User question q , chat history H , trusted vector stores V , language model LLM

Output: Generated answer A , retrieved contexts C

- 1: Authenticate user request
- 2: Check user token balance
- 3: $\text{intent} \leftarrow \text{ClassifyQuery}(q)$
- 4: $\text{temporal_info} \leftarrow \text{ExtractTemporalEntities}(q)$
- 5: $q_norm \leftarrow \text{NormalizeQuestion}(q)$
- 6: $\text{candidate_docs} \leftarrow \text{MultiBrainFAISSRetrieve}(q_norm, V)$
- 7: for each document d in candidate_docs do
- 8: $S_sem \leftarrow \text{SemanticScore}(q_norm, d)$
- 9: $S_temp \leftarrow \text{TemporalScore}(\text{temporal_info}, d)$
- 10: $S_caus \leftarrow \text{CausalScore}(q_norm, d)$
- 11: $S_total \leftarrow \alpha(\text{intent}) \times S_sem$
 $\quad + \beta(\text{intent}) \times S_temp$
 $\quad + \gamma(\text{intent}) \times S_caus$
- 12: end for
- 13: $\text{ranked_docs} \leftarrow \text{SortByScore}(\text{candidate_docs}, S_total)$

```

14: filtered_docs ← DocumentGrader(q, ranked_docs)
15: if filtered_docs is not empty then
16:   C ← filtered_docs
17:   A ← GenerateAnswer(LLM, q, C, H)
18: else
19:   A, C ← TrustAwareSelfLearning(q)
20: end if
21: Save conversation log
22: Update token balance
23: return A, C

```

Algorithm 2. Trust-aware self-learning pipeline

Input: User question q , web search engine W , language model LLM, pending knowledge repository P , dynamic vector store $V_dynamic$, feedback threshold θ

Output: Provisional answer A_web , verified web contexts C_web

```

1: web_results ← SearchWeb(W, q)
2: raw_chunks ← CrawlAndExtract(web_results)
3: relevant_chunks ← FilterRelevantContent(q, raw_chunks)
4: C_web, confidence ← VerifyReliability(LLM, q, relevant_chunks)
5: if confidence is low then
6:   A_web ← "Insufficient evidence"
7:   return A_web,  $\emptyset$ 
8: end if
9: A_web ← GenerateWebBasedAnswer(LLM, q, C_web)
10: SaveToPendingKnowledge(P, q, A_web, C_web)
11: Collect user feedback for q
12: if PositiveFeedback(q)  $\geq \theta$  then
13:   refined_knowledge ← RefineKnowledge(LLM, q, A_web, C_web)
14:   chunks ← ChunkText(refined_knowledge)
15:   embeddings ← GenerateEmbeddings(chunks)
16:   IngestIntoFAISS(V_dynamic, chunks, embeddings)
17:   MarkKnowledgeAsApproved(P, q)
18: end if
19: return A_web, C_web

```

The TALRAG workflow begins when a user submits a historical question. The system first performs authentication and token balance checking to ensure secure and controlled access. The query is then classified into intent types such as factual, temporal, causal, or comparative questions. This classification guides the retrieval process because different historical questions require different evidence priorities.

To support historical reasoning, TALRAG applies a lightweight temporal extraction mechanism to identify years, dynasties, historical periods, and other time-related expressions. These temporal signals are used to guide document retrieval and

ranking, helping the system avoid contexts that are semantically similar but historically misaligned.

The retrieval component is based on multi-source FAISS retrieval. It retrieves candidate documents from trusted historical sources, approved knowledge, and dynamically learned vector memory. Instead of ranking documents only by semantic similarity, TALRAG computes a combined retrieval score:

$$S_{\text{total}} = (\alpha \times S_{\text{semantic}}) + (\beta \times S_{\text{temporal}}) + (\gamma \times S_{\text{causal}})$$

where S_{semantic} measures semantic similarity between the question and the document, S_{temporal} measures chronological consistency, and S_{causal} measures causal alignment with the query. The weights α , β , and γ are adjusted according to the classified query intent. For example, temporal questions assign higher weight to S_{temporal} , while causal questions prioritize S_{causal} .

After retrieval, the ranked documents are processed by an LLM-based document grading module. This module filters noisy, weakly related, or contradictory contexts before answer generation. If sufficient reliable context remains, the LLM generates a grounded answer using the filtered evidence.

If the trusted retrieval pipeline cannot find sufficient reliable knowledge, TALRAG activates the trust-aware self-learning pipeline. In this stage, the system searches external sources and extracts candidate historical evidence. The extracted content is filtered for relevance and checked for reliability before being used to generate a provisional answer. Importantly, knowledge acquired from the web is not directly inserted into the trusted knowledge base. Instead, it is stored in a pending knowledge repository until it receives sufficient positive user feedback.

The central novelty of this framework is the use of user interaction history as a learning signal. When a provisional answer receives enough positive feedback, the automatic learning module refines the verified content, splits it into chunks, generates embeddings, and ingests it into the dynamic FAISS vector store. As a result, future similar questions can retrieve this validated knowledge directly from vector memory.

This design allows TALRAG to continuously improve its historical coverage while reducing the risk of contaminating the trusted knowledge base with unreliable information. Finally, the system records interaction logs, retrieval traces, feedback signals, and token usage for monitoring, resource management, and future analysis.

4 EXPERIMENTS

To evaluate the effectiveness of TALRAG, we conducted a RAGAS-based comparative experiment against the ItihashQA baseline. ItihashQA is a retrieval-augmented historical question answering system originally developed for the Bangladeshi historical context. In this study, its static RAG pipeline is adapted to the Vietnamese historical corpus for controlled comparison. The baseline retrieves the top- k semantically similar chunks from the vector database and directly passes them to the language model for answer generation.

In contrast, TALRAG extends the standard RAG workflow with intent classification, temporal and causal scoring, document grading using an LLM, web

fallback learning, pending knowledge storage, and FAISS ingestion guided by user feedback. This comparison examines whether adaptive retrieval and self learning guided by trust assessment improve both retrieval quality and answer generation quality in Vietnamese historical QA.

The evaluation is conducted on a curated Vietnamese historical corpus consisting of more than 5,000 text chunks derived from historical textbooks, monographs, scholarly articles, and documents covering major Vietnamese dynasties and events. The documents are split into 2,000 character chunks with a 200 character overlap, embedded using the VertexAI text-embedding-004 model, and indexed using FAISS for approximate nearest-neighbor retrieval.

The benchmark dataset contains 30 manually verified Vietnamese historical questions divided into three difficulty levels: Easy, Medium, and Hard. Easy questions focus on factual lookup, such as events, persons, and dates. Medium questions require explanatory reasoning, including causes, consequences, and historical roles. Hard questions involve analytical or comparative reasoning across multiple events, dynasties, or historical periods. Each question is paired with a manually verified ground-truth answer for reference-based evaluation.

The evaluation uses four RAGAS metrics: Faithfulness, Answer Relevancy, Context Precision, and Context Recall. These metrics evaluate both answer generation quality and retrieval quality.

$$\text{Faithfulness} = \frac{\text{supported claims in the answer}}{\text{total claims in the answer}}$$

$$\text{Answer Relevancy} = \frac{1}{N} \sum_{i=1}^N \cos(E(q), E(q_i))$$

$$\text{Context Precision} = \frac{1}{K} \sum_{k=1}^K \text{Precision } k \times \text{rel}(k)$$

$$\text{Context Recall} = \frac{\text{ground} - \text{truth claims supported by retrieved contexts}}{\text{total ground} - \text{truth claims}}$$

In these formulas, $E(q)$ denotes the embedding of the original question, $E(q_i)$ denotes the embedding of a question generated from the answer, and $\text{rel}(k)$ indicates whether the chunk at rank k is relevant.

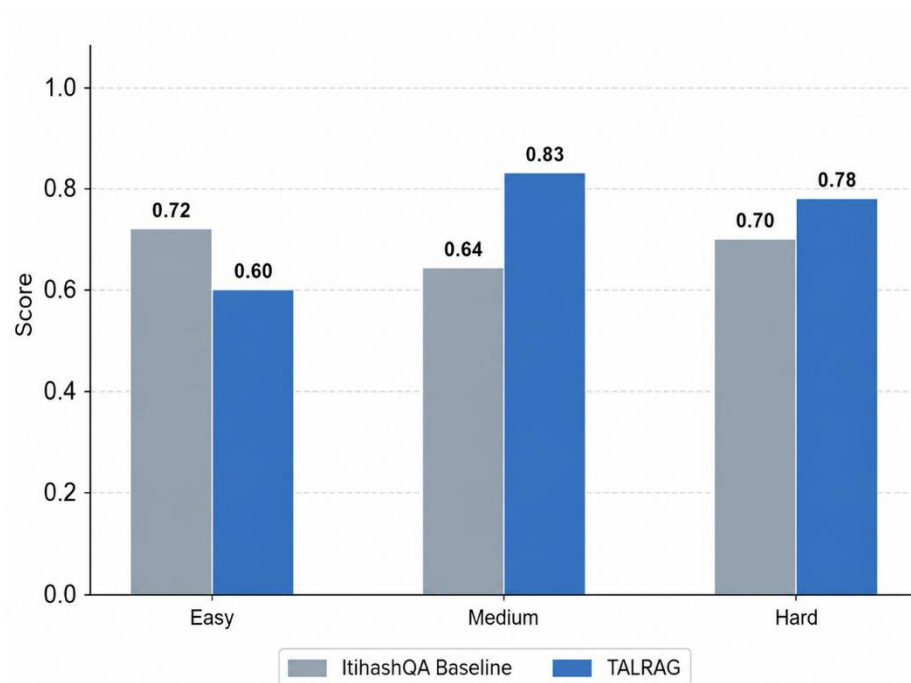


Fig. 2a. Answer Relevancy comparison between ItihashQA baseline and TALRAG.

The largest improvement appears in Answer Relevancy. On Easy questions, TALRAG improves from 0.35 to 0.44. On Medium questions, it increases from 0.18 to 0.79. On Hard questions, TALRAG maintains a score of 0.70, while the baseline drops to 0.09. These results indicate that TALRAG generates answers that are more directly aligned with users' historical questions, especially as question complexity increases.

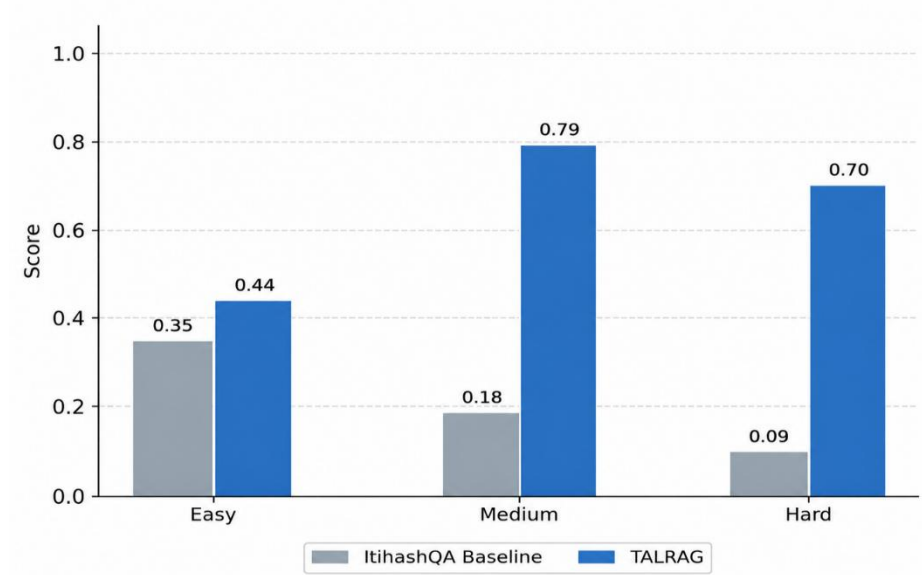


Fig. 2b. Faithfulness comparison between ItihashQA baseline and TALRAG.

For Faithfulness, TALRAG performs better on Medium and Hard questions, with scores of 0.83 and 0.78 compared with the baseline scores of 0.64 and 0.70, respectively. However, on Easy questions, the baseline obtains a higher Faithfulness score of 0.72 compared with TALRAG's 0.60. This suggests that simple factual questions may not always require complex adaptive retrieval, and stricter filtering may sometimes remove useful context for straightforward lookup questions.

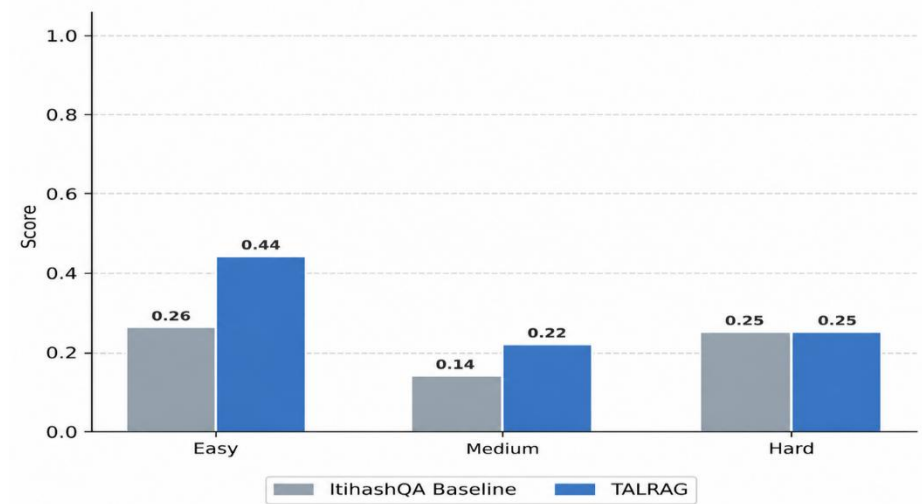


Fig. 2c. Context Precision comparison between ItihashQA baseline and TALRAG.

In terms of Context Precision, TALRAG improves over the baseline across all difficulty levels. On Easy questions, the score increases from 0.26 to 0.44. On Medium questions, it increases from 0.14 to 0.22. On Hard questions, both systems achieve approximately 0.25, with TALRAG showing a slight improvement. These results show that temporal-causal scoring and document grading help retrieve cleaner and more relevant contexts.

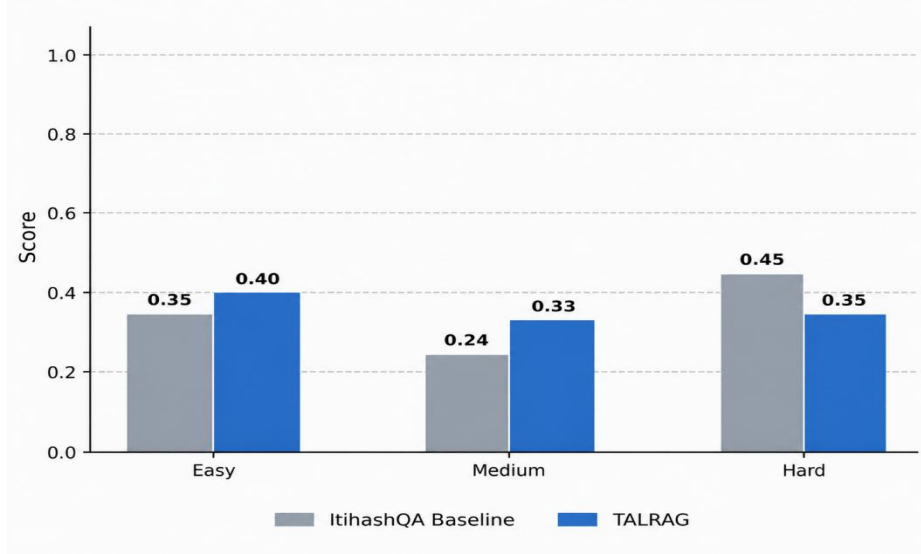


Fig. 2d. Context Recall comparison between ItiHashQA baseline and TALRAG.

For Context Recall, TALRAG improves over the baseline on Easy and Medium questions, increasing from 0.35 to 0.40 and from 0.24 to 0.33, respectively. However, on Hard questions, the baseline obtains a higher score of 0.45 compared with TALRAG’s 0.35. This suggests that the baseline may retrieve broader evidence in some difficult multi-hop questions, while TALRAG prioritizes precision and relevance through stricter filtering.

TALRAG achieves higher average scores than the baseline on all four RAGAS metrics. Faithfulness improves from 0.68 to 0.73, Answer Relevancy improves from 0.21 to 0.64, Context Precision improves from 0.22 to 0.31, and Context Recall improves from 0.35 to 0.36. The largest gain is observed in Answer Relevancy, showing the effectiveness of retrieval based on query intent and adaptive re-ranking.

These results confirm that TALRAG improves the overall quality of Vietnamese historical question answering compared with a static RAG baseline. The improvement is mainly attributed to three components: adaptive retrieval using multiple scores, LLM-based document grading, and the self learning pipeline guided by trust assesment. At the same time, the results show that TALRAG does not outperform the baseline in every individual case. In particular, the lower Easy Faithfulness score and lower Hard Context Recall score suggest that stricter filtering can reduce useful evidence coverage. Future work should investigate adaptive thresholds, larger top-k retrieval for difficult questions, and better precision-recall balancing.

The RAGAS evaluation shows that TALRAG provides a more relevant, context-aware, and RAG architecture that can improve over time for Vietnamese historical QA. By combining temporal and causal retrieval, document grading, and trust-aware self-learning, TALRAG turns static process of retrieval and generation into a continuously evolving historical knowledge system.

5 CONCLUSION

This research introduced TALRAG, a RAG framework with self learning and temporal adaptation for Vietnamese historical question answering. Unlike static RAG systems that mainly rely on semantic similarity and fixed vector databases, TALRAG combines adaptive retrieval, temporal and causal reasoning, document grading, and self learning guided by trust assessment in a unified workflow.

The main contribution of TALRAG is its self learning pipeline guided by trust assessment. When reliable information cannot be retrieved from the trusted vector store, the system acquires supplementary historical evidence from external sources, verifies its reliability, stores it as pending knowledge, and uses user feedback to decide whether it should be ingested into FAISS as reusable vector memory. This mechanism allows TALRAG to learn from user interaction history while reducing the risk of contaminating the knowledge base with unreliable information.

TALRAG demonstrates that RAG can be extended from a static process of retrieval and generation into a continuously evolving historical knowledge system. Future work will focus on improving recall for multi-hop questions, integrating knowledge graphs to model relationships among historical entities, and exploring multimodal retrieval for historical maps, scanned documents, ancient scripts, and artifacts.

REFERENCES

1. Chen, Jiawei, Lin, Hongyu, Han, Xianpei, & Sun, Le. (2023). Benchmarking Large Language Models in Retrieval-Augmented Generation. arXiv preprint arXiv:2309.01414.
2. Devlin, Jacob, Chang, Ming-Wei, Lee, Kenton, & Toutanova, Kristina. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 4171–4186.
3. Ji, Ziwei, Lee, Nayeon, Frieske, Rita, Yu, Tiezheng, Su, Dan, Xu, Yan, Ishii, Etsuko, Bang, Ye Jin, Madotto, Andrea, & Pascual, Damien. (2023). Survey of Hallucination in Natural Language Generation. *ACM Computing Surveys*, 55(12), 1–38.
4. Lewis, Patrick, Perez, Ethan, Piktus, Aleksandra, Petroni, Fabio, Karpukhin, Vladimir, Goyal, Naman, Küttler, Heinrich, Lewis, Mike, Yih, Wen-tau, Rocktäschel, Tim, Riedel, Sebastian, & Kiela, Douwe. (2020). Retrieval-

- Augmented Generation for Knowledge-Intensive NLP Tasks. *Advances in Neural Information Processing Systems (NeurIPS)*, 33, 9459–9474.
5. Nogueira, Rodrigo, & Cho, Kyunghyun. (2019). Passage Re-ranking with BERT. *arXiv preprint arXiv:1901.04085*.
 6. OpenAI. (2023). GPT-4 Technical Report. *arXiv preprint arXiv:2303.08774*.
 7. Riloff, Ellen, & Thelen, June. (2000). A Rule-based Question Answering System for Reading Comprehension Tests. In *Proceedings of the ANLP/NAACL Workshop on Reading Comprehension Tests as Evaluation Resources for NLP*, 13–21.
 8. Thakur, Nandan, Reimers, Nils, Daxenberger, Johannes, & Gurevych, Iryna. (2021). BEIR: A Heterogeneous Benchmark for Information Retrieval Evaluation. *arXiv preprint arXiv:2104.08663*.
 9. Vaswani, Ashish, Shazeer, Noam, Parmar, Niki, Uszkoreit, Jakob, Jones, Llion, Gomez, Aidan N., Kaiser, Lukasz, & Polosukhin, Illia. (2017). Attention Is All You Need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 5998–6008.
 10. Himu, A. D., Azad, M. S., Rahman, R., & Hasan, M. R. (2024). Towards Accurate AI-Driven ItihashQA: A Conversational Question Answering System Applied in Bangladeshi Historical Context. *GitHub repository*. [Online]. Available: <https://github.com/shawmoonazad/ItihashQA>. Accessed: May 14, 2026.
 11. Asai, Akari, Wu, Zeqiu, Wang, Yizhong, Slotnick, Avirup, Sonthalia, Rishi, & Hajishirzi, Hannaneh. (2023). Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. *International Conference on Learning Representations (ICLR)*.
 12. Yan, Shi-Qi, Gu, Jia-Chen, Zhu, Yun-Ze, & Ling, Zhen-Hua. (2024). Corrective Retrieval Augmented Generation. *arXiv preprint arXiv:2401.15884*.
 13. Gao, Yunfan, Xiong, Yun, Gao, Xinyu, Kangxiang, Pan, Jinliu, Bi, Yutong, Dai, Yi, Sun, Jiawei, & Wang, Haofen. (2024). Retrieval-Augmented Generation for Large Language Models: A Survey. *arXiv preprint arXiv:2312.10997*.
 14. Nguyen, Dat Quoc, & Nguyen, Anh Tuan. (2020). PhoNLP: A Joint Overloaded Model for Vietnamese Natural Language Processing. *arXiv preprint arXiv:2011.01544*.
 15. Dhingra, Bhuwan, Cole, Jeremy R., Eisenschlos, Julian Martin, Gillick, Daniel, Eisenstein, Jacob, & Cohen, William W. (2022). Time-Aware Language Models as Temporal Knowledge Bases. *Transactions of the Association for Computational Linguistics*, 10, 257–273.
 16. Ouyang, Long, Wu, Jeffrey, Jiang, Xu, Almeida, Diogo, Wainwright, Carroll, Mishkin, Pamela, Zhang, Chong, Agarwal, Sandhini, Slama, Katarina, Ray, Alex, Schulman, John, Hilton, Jacob, Kelton, Fraser, Miller, Luke, Simens, Maddie, Askell, Amanda, Welinder, Peter, Christiano, Paul, Leike, Jan, & Lowe, Ryan. (2022). Training Language Models to Follow Instructions with Human Feedback. *Advances in Neural Information Processing Systems (NeurIPS)*.
 17. Karpukhin, Vladimir, Oguz, Barlas, Min, Sewon, Lewis, Patrick, Wu, Ledell, Edunov, Sergey, Chen, Danqi, & Yih, Wen-tau. (2020). Dense Passage Retrieval

- for Open-Domain Question Answering. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP).
18. Reimers, Nils, & Gurevych, Iryna. (2019). Sentence-BERT: Sentence Embeddings Using Siamese BERT-Networks. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP).
 19. Edge, Darren, Trinh, Ha, Cheng, Newman, Bradley, Joshua, Chao, Shuo, Mody, Apurva, Truitt, Steven, & Larson, Jonathan. (2024). From Local to Global: A GraphRAG Approach to Query-Focused Summarization. arXiv preprint arXiv:2404.16130.
 20. Johnson, Jeff, Douze, Matthijs, & Jégou, Hervé. (2019). Billion-scale Similarity Search with GPUs. *IEEE Transactions on Big Data*, 7(3), 535–547.
 21. Zheng, Lianmin, Chiang, Wei-Lin, Sheng, Ying, Zhuang, Siyuan, Wu, Zhanghao, Zhuang, Yonghao, Lin, Zi, Li, Zhuohan, Li, Dacheng, Xing, Eric, Zhang, Hao, & Stoica, Ion. (2023). Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. arXiv preprint arXiv:2306.05685.
 22. Google. (2023). NotebookLM. [Online]. Available: <https://notebooklm.google/>. Accessed: May 14, 2026.
 23. Google. (2024). Gemini Gems. [Online]. Available: <https://gemini.google.com/>. Accessed: May 14, 2026.